# Group I Row III: why the logarithms carry a modulus

step-by-step derivation

## 1 Setup

The Fourier transforms (i3)–(i5) are principal values, e.g.

$$\int \frac{d^2k}{2\pi}\frac{d^2p}{2\pi}\,\frac{e^{ik\cdot X+ip\cdot Z}}{\xi k^2-p^2} = \mathrm{PV}\frac{1}{\xi Z^2-X^2}. \tag{1.1}$$

In the cross section of Group I Row III this gives the denominator $p^+(x-z)^2-k^+(x-w)^2$, which is then integrated over $p^+$. Write

$$A\equiv(x-z)^2>0, \qquad B\equiv(x-w)^2>0, \qquad u(p^+)\equiv p^+A-k^+B, \tag{1.2}$$

so every logarithm of the row comes from

$$I(p_1,p_2)=\int_{p_1}^{p_2}dp^+\,\frac{A}{p^+A-k^+B}, \tag{1.3}$$

with $(p_1,p_2)=(k^+,\vee)$ (part 1 line 2, part 2), $(0,k^+)$ (part 1 line 3, part 3a), or $(0,\vee)$ (part 1 line 4).

## 2 Derivation

**Step 1: the pole.** $u(p^+)$ is linear in $p^+$ with slope $A>0$. It vanishes at

$$p_0^+=k^+\frac{B}{A}=k^+\frac{(x-w)^2}{(x-z)^2}. \tag{2.1}$$

This is the singular surface $\xi Z^2=X^2$ of (1.1), written in the $p^+$ variable.

**Step 2: the prescription is inherited from the Fourier transform.** Because (1.1) is a principal value, so is the $p^+$ integrand. Since $u$ is linear in $p^+$,

$$\mathrm{PV}\frac{1}{p^+A-k^+B}=\frac{1}{A}\,\mathrm{PV}\frac{1}{p^+-p_0^+}, \tag{2.2}$$

and the symmetric excision $|u|<\delta$ is exactly the symmetric excision $|p^+-p_0^+|<\delta/A$. Nothing new is assumed: the $p^+$ integral must be a principal value.

**Step 3: the real antiderivative.** For $u\neq 0$,

$$\frac{d}{dp^+}\log|p^+A-k^+B|=\frac{A}{p^+A-k^+B}, \tag{2.3}$$

since $\frac{d}{du}\log|u|=\frac{1}{u}$ for $u>0$ *and* for $u<0$. This is the antiderivative that is real on both sides of the pole.

**Step 4a:** $p_0^+ \notin (p_1, p_2)$**.** No singularity; $u(p_1)$ and $u(p_2)$ have the same sign:

$$I = \log \frac{p_2 A - k^+ B}{p_1 A - k^+ B} = \log \left| \frac{p_2 A - k^+ B}{p_1 A - k^+ B} \right|. \tag{2.4}$$

The ratio is positive and the modulus changes nothing.

**Step 4b:** $p_0^+ \in (p_1, p_2)$**.** Now $u(p_1) < 0 < u(p_2)$ and

$$\begin{aligned} I &= \lim_{\delta \to 0} \left[ \int_{p_1}^{p_0^+ - \delta} + \int_{p_0^+ + \delta}^{p_2} \right] \frac{A\, dp^+}{p^+ A - k^+ B} \\ &= \lim_{\delta \to 0} \Big[ \log |u(p_0^+ - \delta)| - \log |u(p_1)| + \log |u(p_2)| - \log |u(p_0^+ + \delta)| \Big] \\ &= \lim_{\delta \to 0} \Big[ \log(A\delta) - \log |p_1 A - k^+ B| + \log |p_2 A - k^+ B| - \log(A\delta) \Big], \end{aligned} \tag{2.5}$$

using $|u(p_0^+ \mp \delta)| = A\delta$. The two $\log(A\delta)$ cancel exactly (this is where the *symmetric* excision matters), and

$$\boxed{\mathrm{PV} \int_{p_1}^{p_2} \frac{dp^+\, A}{p^+ A - k^+ B} = \log \left| \frac{p_2 A - k^+ B}{p_1 A - k^+ B} \right|} \tag{2.6}$$

Here the ratio inside the modulus is *negative*.

**Step 5: what the naive** log(**negative**) **was.** Without a prescription one writes $\log \frac{p_2 A - k^+ B}{p_1 A - k^+ B} = \log(\text{negative}) = \log|\cdot| \pm i\pi$. That $\pm i\pi$ is the $\delta$-function part of a Feynman prescription:

$$\frac{1}{u \pm i\epsilon} = \mathrm{PV} \frac{1}{u} \mp i\pi\, \delta(u), \qquad \int_{p_1}^{p_2} dp^+ A\, \delta(p^+ A - k^+ B) = \begin{cases} 1, & p_0^+ \in (p_1, p_2) \\ 0, & \text{otherwise} \end{cases} \tag{2.7}$$

so $\int_{p_1}^{p_2} \frac{dp^+ A}{p^+ A - k^+ B \pm i\epsilon} = \log \left| \frac{p_2 A - k^+ B}{p_1 A - k^+ B} \right| \mp i\pi$ when $p_0^+$ is inside. The energy denominator is $+i\epsilon$ in the amplitude and $-i\epsilon$ in the complex-conjugate amplitude; in the cross section (amplitude $\times$ c.c. + c.c.) the two $i\pi\delta$ pieces cancel. That is why (1.1) is a principal value and why $d^3N$ is real. The correct result is therefore the real part, $\log|\cdot|$, with no $i\pi$.

## 3 Applied to Row III

| Terms | $(p_1, p_2)$ | Result (2.6) | negative argument iff |
|---|---|---|---|
| part 1 line 2, part 2 | $(k^+, \vee)$ | $\log \left\| \frac{\vee (x-z)^2 - k^+ (x-w)^2}{k^+ \left[ (x-z)^2 - (x-w)^2 \right]} \right\|$ | $k^+ < p_0^+ < \vee$ |
| part 1 line 3, part 3a | $(0, k^+)$ | $\log \left\| \frac{(x-z)^2 - (x-w)^2}{(x-w)^2} \right\|$ | $0 < p_0^+ < k^+$ |
| part 1 line 4 | $(0, \vee)$ | $\log \left\| \frac{\frac{\vee}{k^+} (x-z)^2 - (x-w)^2}{(x-w)^2} \right\|$ | $0 < p_0^+ < \vee$ |

For part 1 line 4, dividing numerator and denominator by $\frac{\vee}{k^+}$,

$$\log \left| \frac{\frac{\vee}{k^+} (x-z)^2 - (x-w)^2}{(x-w)^2} \right| = \log \frac{\vee}{k^+} + \log \left| \frac{(x-z)^2 - \frac{k^+}{\vee} (x-w)^2}{(x-w)^2} \right|, \tag{3.1}$$

whose second log is your collaborator's $\log \frac{\frac{k^+}{\vee}(x-w)^2 - (x-z)^2}{(x-w)^2}$ (up to the overall sign inside the modulus, which the modulus removes). Its argument is negative exactly when $p_0^+ < \vee$, i.e. for almost every transverse configuration when $\vee$ is large: that is the negative log your collaborator found. With the modulus it is well defined. The split-off $\log \frac{\vee}{k^+}$ and $\log \frac{k^+}{\Lambda}$ have positive arguments and are unchanged.

## 4 Numerical check

`check.py` ($k^+ = 1$; scipy's Cauchy-weight quadrature for the PV; the $+i\epsilon$ integral evaluated as $\log(u_2 + i\epsilon) - \log(u_1 + i\epsilon)$ with $\epsilon = 10^{-3}$):

| $A$ | $B$ | $(p_1, p_2)$ | $p_0^+$ | PV integral | $\log\lvert\cdot\rvert$ of (2.6) | $+i\epsilon$: Im/$\pi$ |
|---|---|---|---|---|---|---|
| 0.7 | 1.9 | $(0, 10)$ | 2.714 (in) | 0.9873866536 | 0.9873866536 | $-0.9998$ |
| 0.7 | 1.9 | $(1, 10)$ | 2.714 (in) | 1.4469189829 | 1.4469189829 | $-0.9997$ |
| 1.3 | 0.4 | $(0, 1)$ | 0.308 (in) | 0.8109302162 | 0.8109302162 | $-0.9989$ |
| 0.5 | 0.2 | $(1, 10)$ | 0.400 (out) | 2.7725887222 | 2.7725887222 | 0 |

The PV equals $\log\lvert\cdot\rvert$ in every case; the $+i\epsilon$ version has the same real part and an extra $-i\pi$ (Im/$\pi \to -1$ as $\epsilon \to 0$) only when the pole is inside the range, which the conjugate amplitude cancels.